

Some review questions - 8th week

Q1 Consider the linear system of equations given in matrix form by:

$$\begin{bmatrix} I_r & F \\ 0_{(m-r) \times r} & 0_{(m-r) \times (n-r)} \end{bmatrix} \begin{bmatrix} x_p \\ x_F \end{bmatrix} = \begin{bmatrix} b_{r \times 1} \\ c_{(m-r) \times 1} \end{bmatrix}.$$

For each block form below, fill in the table with the possible number of solutions. Specify, when relevant, how the value of $c_{(m-r) \times 1}$ affects the number of solutions.

Block type	$[I_r]$ $r = m = n$	$[I_r \ F]$ $r = m < n$	$\begin{bmatrix} I_r \\ 0 \end{bmatrix}$ $r = n < m$	$\begin{bmatrix} I_r & F \\ 0 & 0 \end{bmatrix}$ $r < m, \ r < n$
Possible nb of solutions				

Q2 Give the definitions of full row rank, full column rank, and full-rank matrices.

Q3 After performing row operations, determine whether the following matrix has full row rank, full column rank, and/or full rank:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 3 & 5 \\ 3 & 3 & 5 \end{bmatrix}$$

Q4 Given two (possibly equal) integers $n, m \in \mathbb{N}$, a vector $b \in \mathbb{R}^m$ and a matrix $A \in \mathbb{R}^{m \times n}$ such that $\text{rank}(A) = n$, say how many solutions the equation $Ax = b$ can have.

Q5 Given two (possibly equal) integers $n, m \in \mathbb{N}$, a vector $b \in \mathbb{R}^m$ and a matrix $A \in \mathbb{R}^{m \times n}$ such that $\text{rank}(A) = m$, say how many solutions the equation $Ax = b$ can have.

Q6 Give the definition and notation of orthogonality for (i) two vectors $u, v \in \mathbb{R}^n$, (ii) a vector $u \in \mathbb{R}^n$ and a subspace $U \subset \mathbb{R}^n$, (iii) two subspaces $U, V \subset \mathbb{R}^n$.

Q7 Express the set of all vectors orthogonal to $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \in \mathbb{R}^3$. Is this set a subspace of $\mathbb{R}^3$?

Q8 Give the definition of the orthogonal complement.

Q9 Express the orthogonal complement of $\text{Span} \left(\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\} \right)$ as a span.

Q10 Given a subspace $S \subset \mathbb{R}^m$, is $S^\perp$ also a subspace? Given a basis $\{u_1, \dots, u_r\}$ of S and a basis $\{v_1, \dots, v_l\}$ of $S^\perp$, provide a relation between r and l . What can be said about the set of vectors $\{u_1, \dots, u_r, v_1, \dots, v_l\}$?

Q11 Given a matrix $A \in \mathbb{R}^{m \times n}$, give the definition of the sets $N(A)$, $N(A^T)$, $\text{Row}(A)$, and $C(A)$. Specify the linear spaces in which each of these sets is contained.

Q12 Given a matrix $A \in \mathbb{R}^{m \times n}$, give two relations involving the sets $N(A)$, $N(A^T)$, $Row(A)$, and $C(A)$, and prove one of them.

Q13 Given a matrix $A \in \mathbb{R}^{m \times n}$, what is the relation between $Row(A)$ and $C(A^T)$? What about $Row(A^T)$ and $C(A)$?

Q14 State the rank-nullity theorem.

Q15 Using a method covered in class, explain how you would estimate a reasonable salary for Tom based on the given data. You can introduce the variable names a, b, s for *Score*, *Assist*, and *Salary*, respectively. Numerical computation is not required.

	Score	Assist	Salary
Paul	16	10	35
Harden	25	8	43
Capela	18	3	16
Tom	12	14	???

Q16 Express the least squares problem, ensuring that you clearly define all objects involved in the setting.

Q17 Given $b \in \mathbb{R}^m$ and $A \in \mathbb{R}^{m \times n}$, determine whether the following two problems $(P1)$ and $(P2)$ are equivalent (in the sense that they have the same set of solutions). If the answer is no, give a counterexample; if the answer is yes, prove it.

$$(P1) : \min_x \|Ax - b\|$$

$$(P2) : \min_x \|Ax - b\|^2.$$

Q18 Given a vector $b \in \mathbb{R}^m$ and a matrix $A \in \mathbb{R}^{m \times n}$, express the least squares problem $\min_x \|Ax - b\|$ as the minimization of a sum of squares of residual components.

Q19 Given a vector $b \in \mathbb{R}^m$ and a matrix $A \in \mathbb{R}^{m \times n}$, express the least squares problem $\min_x \|Ax - b\|$ as the minimization of the distance between b and a linear space that you will introduce.

Q20 Given a vector $b \in \mathbb{R}^m$, a subspace $S \subset \mathbb{R}^m$, and a vector $p \in S$, give a necessary and sufficient condition on p , b , and S under which:

$$\forall z \in S: \quad \|b - p\| \leq \|b - z\|.$$

Q21 Given a vector $b \in \mathbb{R}^m$ and a subspace $S \subset \mathbb{R}^m$, give the definition of the projection p of b onto S .

- Q22** What can be said about a vector $x \in \mathbb{R}^n$ if for all $u \in \mathbb{R}^n$, $u^T x = 0$? Prove your result. Is the converse true?
- Q23** Given a vector $b \in \mathbb{R}^m$ and a matrix $A \in \mathbb{R}^{m \times n}$, give a necessary and sufficient condition, in the form of a linear equation involving A , y , and b , under which y is a solution to the minimization problem $\min_x \|Ax - b\|$ (no proof needed). What is this equation called?
- Q24** State the normal equation. Does it have a solution? (No proof needed.)
- Q25** Given a vector $b \in \mathbb{R}^m$ and a matrix $A \in \mathbb{R}^{m \times n}$, prove that $b - Ay \perp C(A)$ is equivalent to $A^T Ay = A^T b$. You may use the result “if $\forall u \in \mathbb{R}^n$, $u^T x = 0$, then $x = 0$ ”.